

SEASONAL DYNAMISM OR MODEL REALISM: WHAT DRIVES BETTER PREDICTIONS OF LANDSCAPE CONNECTIVITY?

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Data Statement: All covariate data and analysis code are provided through Zenodo. Species data
supporting this research are sensitive and not available publicly. Detailed GPS data of dispersing African
wild dogs are available to qualified researchers from the University of Zurich by contacting Dr. David
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Zurich (david.hofmann2@uzh.ch). Sharing of data is contingent on approval of the request by
Botswana Predator Conservation and University of Zurich.

Connectivity modeling is central to conservation science, yet most studies assume static landscapes and fixed species-habitat preferences, thus overlooking seasonal dynamics. Additionally, many rely on simplistic species-habitat associations rather than realistic movement models that mechanistically capture animal behavior. These limitations risk masking key ecological processes, leading to poor agreement between predicted and observed movements.

We propose and employ a conceptual framework illustrating that seasonality can be incorporated at three stages of the connectivity modeling pipeline: (1) when extracting spatial covariates for model fitting, (2) when specifying the selection model, and (3) when making predictions. This yields several model configurations varying in seasonal dynamism. Further, we compare the performance of alternative models that vary in their degree of biological realism, ranging from simplistic habitat-based models to realistic formulations that consider both environmental and behavioral drivers of movement.

Using GPS data from dispersing African wild dogs (*Lycaon pictus*) in the seasonally dynamic Okavango Delta, we assess whether accounting for seasonality improves predictions of dispersal trajectories and connectivity. We fit integrated step-selection functions and predict connectivity using individual-based dispersal simulations while explicitly considering seasonal dynamism in both environmental covariates and the species' habitat and movement preferences. Furthermore, predictions from a simplistic habitat-based model are contrasted with those from a more mechanistic and biologically realistic model enabled by individual-based simulations.

We evaluate predictive performance through cross-validation across seasonal configurations and levels of model realism. We hypothesized that incorporating seasonality and using a realistic movement model would improve predictions. Contrary to expectations, seasonality modestly improved predictions when using a simplistic model, but had no added benefit when using the realistic model, which already performed well.

Our findings offer a framework for integrating seasonality into connectivity studies and highlight the greater value of realistic movement models over simplistic ones. While seasonal dynamics are important, their benefits may be limited when biologically informed models are already in place. Connectivity studies should therefore prioritize mechanistic approaches to better capture species movement and landscape use.

57 INTRODUCTION

58 Landscape connectivity is defined as the degree to which the landscape facilitates or impedes
59 movement among habitat patches (Taylor et al., 1993) and is a central prerequisite for maintaining
60 biodiversity (Fahrig, 2003). Improved connectivity facilitates dispersal (Baguette et al., 2013; Doerr et
61 al., 2011), which in turn promotes genetic diversity (Frankham et al., 2002; Perrin & Mazalov, 2000)
62 and the colonization of vacant habitats (Hanski, 1999; MacArthur & Wilson, 2001). Due to its beneficial
63 impacts on metapopulation dynamics, restoring connectivity is among the most frequently
64 recommended strategies to conserve biodiversity and to promote resilience against climate change
65 (Heller & Zavaleta, 2009; Rudnick et al., 2012). Quantifying connectivity and identifying critical dispersal
66 corridors have therefore become crucial tasks in conservation science (Heller & Zavaleta, 2009;
67 Hofmann et al., 2021; Keeley et al., 2019; Rudnick et al., 2012). Reliable estimation of connectivity is
68 thus paramount for a targeted allocation of scarce conservation funds. Two often-overlooked factors
69 that can hinder our ability to predict connectivity are the neglect of seasonal changes in landscape
70 characteristics and the use of simplistic models that fail to capture the behavioral nuances of
71 movement (Hofmann et al., 2023; Osipova et al., 2019; Zeller, Wattles, et al., 2020).

72 Seasonality can impact functional connectivity both through spatio-temporal variation in the landscape
73 itself, or through temporal variation in species' preferences towards prevailing conditions (Forrest et
74 al., 2024; Mui et al., 2017; Simpkins & Perry, 2017; Zeller, Wattles, et al., 2020). In ecosystems that
75 experience alternations between wet and dry seasons, for example, the onset of the rainy season
76 initiates distinct "green-up" waves, which affect the availability of food resources for herbivores and,
77 subsequently, shape their movements (Merkle et al., 2016). The variation in environmental conditions,
78 for instance, drives herbivore migrations across massive spatial scales, resulting in short-lived
79 movement corridors between otherwise disconnected habitats (Naidoo et al., 2016; Serneels &
80 Lambin, 2001). Alternatively, seasonality can affect functional connectivity via temporal changes in a

species' habitat preferences. Amphibians, for example, require both aquatic and terrestrial habitats, but their preference for one habitat over the other heavily depends on the season (Baldwin et al., 2006). Although such seasonal intricacies are likely to play a fundamental role in many ecosystems, they are only rarely considered in connectivity studies in an explicit manner. In fact, most connectivity studies represent their study system by a static set of environmental layers and assume their focal species to exhibit a fixed set of preferences (e.g., Elliot et al., 2014; Abrahms et al., 2017; but see Forrest et al., 2024; Mui et al., 2017). Overcoming these limitations requires viewing connectivity as a dynamic process and modeling it accordingly (Zeller, Lewison, et al., 2020; Zeller, Wattles, et al., 2020).

A second major challenge in accurately predicting connectivity arises from the use of simplistic models that fail to capture the complexity of animal movement. Traditional approaches to study functional connectivity generally follow a three-step pipeline: (1) compiling presence or movement data for the focal species, alongside spatial covariate layers that influence connectivity, (2) fitting a selection model to estimate species preferences for environmental features and (3) generating a resistance (or permeability) surface based on inferred preferences, which is then used as input to a connectivity model (Diniz et al., 2019; Zeller et al., 2012). One downside to such resistance-based approaches is that models are limited to variables that can be represented as spatial layers. This prevents scientists from including biologically relevant, yet non-spatial effects, such as, for example, the effect of time of day on movement and habitat preferences. More recently, an increasingly popular alternative has been to bypass the resistance surface and simulate individual movement directly from the selection model, thus inferring connectivity from emergent movement patterns (Allen et al., 2016; Hauenstein et al., 2019; Hofmann et al., 2023; Kanagaraj et al., 2013; Unnithan Kumar et al., 2022, 2022; Zeller, Wattles, et al., 2020). This paradigm shift has been accelerated by the rise of integrated step-selection functions (iSSFs), which allow jointly modeling an animal's movement kernel and habitat selection (Avgar et al., 2016; Fieberg et al., 2021) and permit simulating movement directly from a fitted model (Avgar et al.,

2016; Signer et al., 2017). Unlike resistance-based approaches, iSSF-based simulations are not limited to variables that can be represented as raster layers, rather iSSFs offer greater realism by capturing non-spatial, dynamic factors—such as directional persistence, circadian rhythms, and interactions between movement behavior and environmental variables (Forrest et al., 2024; Hofmann et al., 2023; Potts & Börger, 2023; Signer et al., 2024).

Accounting for seasonality in connectivity analyses is non-trivial, as it can enter the modeling pipeline at three distinct stages (Figure 1Error! Reference source not found.a). (1) When compiling spatial covariates (e.g., vegetation, water cover), researchers can either use a single layer per covariate (representing a static or averaged snapshot) or a series of layers that capture seasonal variation (Figure 1a, Stage 1). Historically, static representations have been the norm (e.g., Brennan et al., 2020; Elliot et al., 2014). However, advances in remote sensing and increased access to petabytes of landscape data (Rumiano et al., 2020; Toth & Józków, 2016) have made high-resolution spatiotemporal layers more accessible, enabling more dynamic representations of study systems (Forrest et al., 2024; Kaszta et al., 2021; Osipova et al., 2019). (2) When fitting a selection model, species' habitat preferences can either be assumed constant by pooling all data or modeled separately by season to account for seasonal preferences (Figure 1a, Stage 2; e.g., Cushman and Lewis 2010; Fortin et al. 2005; Manly et al. 2007; Zeller, Wattles, et al. 2020). For instance, Chetkiewicz and Boyce (2009) partitioned their data by season to estimate seasonal habitat preferences for pumas (*Puma concolor*) and grizzly bears (*Ursus arctos*). (3) Finally, one must decide whether to predict connectivity for “average” environmental conditions or for distinct seasons (Figure 1a, Stage 3). Using resistance-based models, the latter approach requires repeatedly applying connectivity models to seasonally updated resistance surfaces (e.g., Kaszta et al., 2021; Osipova et al., 2019; Zeller, Wattles, et al., 2020). This is demonstrated *ad extremum* by Kaszta et al. (2021), who predicted dynamic connectivity for African elephants based on monthly updated resistance surfaces (*Loxodonta africana*). Simulations from iSSFs allow to more

129 elegantly incorporate seasonality, as simulations eliminate the need for seasonally updated resistance
130 surfaces. Instead, one can directly incorporate seasonality by adjusting both the environmental
131 features and the species' habitat preferences as simulated individuals move. However, such an
132 approach has not yet been employed. Regardless of the method used, integrating seasonality into
133 connectivity analyses remains analytically and computationally demanding (Bishop-Taylor et al., 2018),
134 raising the question of when, if, and to what extent it should be prioritized.

135 To study the importance of seasonal dynamism and a more realistic representation of movement when
136 predicting connectivity, we use data from a free-living population of the endangered African wild dog
137 (*Lycaon pictus*) inhabiting the Okavango Delta ecosystem in northern Botswana (McNutt, 1996). The
138 system is well suited to address our objectives, for the Okavango Delta ecosystem experiences
139 significant seasonal variation. African wild dogs are pack-living carnivores that are primarily active
140 during the cooler morning and evening hours (Rasmussen & Macdonald, 2012), as well as during
141 moonlit nights (Cozzi et al., 2012). Their nocturnal activity is strongly mediated by moonlight
142 availability, as wild dogs lack the highly developed night vision typical of felids. Elevated ambient
143 temperatures are associated with shorter activity periods, as wild dogs typically rest during the hottest
144 parts of the day (Rabaiotti & Woodroffe, 2019). Upon reaching sexual maturity, individuals born into a
145 pack disperse in same-sex coalitions to find mates and a territory to settle (McNutt, 1996). Euclidean
146 distances moved by dispersers range from a few dozen to 500 km, with some coalitions covering
147 several hundred kilometers within only a few days (Cozzi et al., 2020, 2023; Davies–Mostert et al.,
148 2012; Masenga et al., 2016; Sandoval-Seres et al., 2022). Studies of habitat-selection show that
149 dispersing wild dogs avoid venturing into large water bodies, prefer moving along it, prefer moving
150 across open grassland or shrubs, yet avoid areas dominated by humans and densely covered by forests
151 (Hofmann et al., 2021; O'Neill et al., 2020). Because functional landscape connectivity is established
152 through dispersal rather than resident space use, we explicitly focus on dispersal movements, which

153 differ fundamentally from resident movements and habitat selection, capturing the long-distance
154 processes most relevant to connectivity (Diniz et al., 2019; Elliot et al., 2014).

155 Our aims are to (1) elaborate a framework for incorporating seasonality into connectivity analyses, (2)
156 evaluate whether including seasonality improves the predictive performance of dispersal models, and
157 (3) assess the value of more realistic representations of movement when predicting connectivity. To
158 achieve this, we compare the predictive performance of six seasonal configurations (Figure 1a) and
159 two movement models – one simplistic and one realistic (Figure 1b). We compile an extensive set of
160 remote sensing data that accurately represent seasonal landscape changes across the Okavango Delta
161 ecosystem and combine these data with dispersal GPS data from 33 dispersing wild dog coalitions in
162 northern Botswana. We then apply k-fold cross-validation for case-control studies to assess predictive
163 efficacy under each configuration and model type. Lastly, we use iSSF-based simulations to estimate
164 connectivity under different degrees of dynamism and under both a simplistic and realistic model. We
165 hypothesize that habitat selection and movement behavior of dispersing wild dogs differ between
166 seasons and that increasing seasonal dynamism improves the predictive efficacy of the simulation
167 model (Figure 1c). We also anticipate that a realistic movement model yields markedly better
168 predictions than a simplistic one.

169 **METHODS**

170 We used the R programming language (R Core Team, 2025) for all data preparation and analyses. We
171 performed spatial data manipulation using the terra (Hijmans et al., 2024) and spatstat (Baddeley et
172 al., 2015) packages. We generated figures using the ggplot2 (Wickham et al., 2024) and ggpubr
173 (Kassambara, 2024) packages.

174 STUDY AREA

175 The study area comprised the Okavango Delta ecosystem in northern Botswana (centered at 24°30'E
176 and 20°42'S at an elevation of approx. 950 m) and extended to parts of eastern Namibia and western
177 Zimbabwe, encompassing an extent of 160,000 km² (Figure 2Error! Reference source not found.). The Okavango
178 Delta is a flood-pulse driven mosaic of patchy woodlands, permanent swamps, and seasonally flooded
179 grasslands that lie within the otherwise dry and sandy Kalahari Basin (Mendelsohn et al., 2010;
180 Ramberg et al., 2006; Wilson & Dincer, 1976). Precipitation across the study area varies considerably
181 between seasons, ranging from 0 mm/month during the dry season (from 15 April to 15 October) to
182 140 mm/month during the wet season (from 15 October to 15 April), totaling to 600 mm across an
183 average year (Figure S1a). Daily maximum above-ground temperature fluctuates between 7°C during
184 the dry winter months and 38°C during the wet summer months (Figure S1b). The vegetation in the
185 study area is mainly composed of mopane (*Colophospermum mopane*) forest, mixed woodland acacia-
186 dominated (*Acacia spp.*), and grassland. Substantial vegetation green-up (e.g., plant and shoots
187 growth, leaf production, and grass greening) starts with a delay of some weeks after the onset of the
188 first rains in the wet season. The normalized difference vegetation index (NDVI) therefore depicts a
189 smoothed and lagged response to precipitation patterns across the study area (Figure S1c). The yearly
190 flood-cycle of the Okavango Delta is predominantly driven by rainfalls in the Angolan highlands, where
191 water is collected and channeled through the Cubango and Cuito rivers into the Okavango Delta
192 (Gumbricht et al., 2004; McCarthy et al., 2003; Mendelsohn et al., 2010). Because water only slowly
193 descends from the catchment areas in Angola into the Delta's tributaries, the flood is out of sync with
194 local rainfalls and reaches its maximum extent between August and September, i.e. during peak dry
195 season (Wolski et al. 2017; Figure S1d). While the extent of large-bodied rivers and floodplains is
196 determined by precipitation in Angola, the emergence of smaller, ephemeral water-bodies (i.e., pans)
197 is driven by local precipitation during the wet season. 62% of the landscape in the study area form part

of formally protected areas (i.e., National Parks, Game Reserves, and Wildlife Management Areas), such that human impact is low and largely limited to settlements along the western part of the Okavango Delta and the city of Maun at the Delta's southern tip (Figure 2). Landscapes outside protected areas in Zimbabwe, however, are more heavily influenced by humans, mainly through agricultural fields and human settlements.

GPS DATA

Between 2011 and 2022, we collected GPS data on 33 dispersing wild dog coalitions (16 female coalitions, 17 male coalitions; Figure 2b, Figure S2, and Figure S3). We programmed GPS satellite collars to record GPS locations at 01:00, 05:00, 09:00, 17:00, and 21:00. In terms of wild dog activity, the eight-hour window between 09:00 and 17:00 can be considered comparable to a four-hour window during the rest of the day, as wild dogs generally rest between 11:00 and 15:00 (Cozzi et al., 2012; Hayward & Slotow, 2009), leaving approximately four hours of activity. The collected GPS data were regularly transmitted to a base-station via Iridium satellite, thereby allowing dispersing individuals to be monitored, even if they ventured across national borders. In total, we obtained 5,940 locations during dispersal, with an average of 198 ± 239 locations per coalition. Occasionally, the acquisition of a GPS location failed (success rate = $93 \pm 8\%$), resulting in slight deviations from the aspired four-hourly schedule. We distinguished dispersal periods from residency using the net-squared displacement (NSD) metric (Börger & Fryxell, 2012) in combination with direct field observations. NSD was calculated as the squared Euclidean distance between each GPS relocation and the center of the coalition's natal home range. Dispersal was defined as the period from emigration from the natal range (sustained increase in NSD) until settlement, indicated by stabilization of NSD, and was corroborated by ground assessments confirming whether individuals left their natal packs (details in Cozzi et al., 2020; Hofmann et al., 2021).

221 COVARIATES

222 We represented the physical landscape through which dispersers moved by a suite of spatial covariates
223 known to influence wild dog movements during dispersal. We broadly categorized these into
224 descriptors of (1) landscape characteristics, (2) climatic conditions, and (3) anthropogenic features
225 (Table 1; see Hofmann et al. 2021; 2023). Besides spatial covariates, we also prepared a series of
226 covariates relating to (4) light intensity (Table 1; see Cozzi et al. 2012) to appropriately render
227 seasonality in each of the spatial covariates (1–3), we downloaded them at the highest spatial and
228 temporal resolutions available. That is, for each spatial covariate, we obtained a stack of raster layers
229 spanning the years 2011 to 2022, for which GPS data on dispersing wild dogs were available. Depending
230 on the covariate, this resulted in differing spatial and temporal resolutions (Table 1). Additionally, for
231 each spatial covariate, we also generated a static layer, representing “average” conditions across the
232 entire duration of the study. For this, we flattened each covariate stack into a single layer, thus
233 removing seasonal and yearly variation from the spatial data entirely. For continuous covariates, we
234 achieved this by averaging conditions across all collected layers, whereas for categorical (binary) layers,
235 we identified areas that were covered by the respective category in at least 50% of all layers. Using the
236 same aggregation techniques, we computed covariate stacks representative of a typical year. That is,
237 instead of removing seasonality by flattening across the entire range of dates, we flattened stacks
238 across years, thereby eliminating year-specific effects while retaining seasonal variation. For example,
239 the floodmap for day one of the typical year was computed by averaging all floodmaps collected on
240 day one across different years. These latter layers were ultimately used in the dispersal simulation,
241 where we needed spatial layers that depicted seasonal dynamism across a typical year. In summary,
242 we prepared each covariate statically, dynamically for the entire range of dates considered, and
243 dynamically for an average year. Static covariates represent a single temporally invariant value per
244 pixel, whereas dynamic covariates consist of pixel-level time series capturing temporal variation.

245 Covariates were selected a priori based on our ecological understanding of the focal species, and we
246 did not remove predictors based on collinearity diagnostics (Morrissey & Ruxton, 2018; Vanhove,
247 2021). Further details on the covariate preparation are provided in Appendix A3 and by Hofmann et al.
248 (2021).

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Table 1: Covariates that were used in this study, including information on their temporal and spatial resolutions, as well as on the avenue through which the respective data were accessed or downloaded. Download methods printed in *typewriter* font indicate the use of an R-package that is available through CRAN or GitHub.

Category	Variable	T y p e	Description	Temporal Resolutio n	Spatial Resolutio n (m)	Source	Downlo ad Method	Refe renc e
Landscape Characteris tics	Trees	C	Percentage tree cover	1 Year	250	MODIS MOD44B	RGIST ools	(1)
	Shrubs / grasslan d	C	Percentage non-tree vegetation	1 Year	250	MODIS MOD44B	RGIST ools	(1)
	NDVI	C	Normalized difference in vegetation index	16 Days	250	MODIS MOD13Q1	rgee	(2)
	Rivers	B	Presence of rivers	Static	90	MERIT Hydro	website	(3)
	Perman ent water	B	Presence of permanent water bodies	Static	30	Globeland 30	Website	(4)
	Floodwa ter	B	Presence of seasonal flood water	8 Days	500	MCD34A4	flood mapr	(5)
	Distance to water	C	Distance to nearest river, permanent water, or seasonal flood (m)	8 Days	500	MCD43A4	flood mapr	(5)
	Pans	C	Presence of ephemeral pans containing water	5/10 Days	10	Sentinel 2	sen2r	(6)
	Distance to pans	C	Distance to nearest ephemeral pan containing water (m)	5/10 Days	10	Sentinel 2	sen2r	(6)
Climate	Temper ature	C	2-m above-ground temperature (Celsius)	4 Hours	1,000	ERA5	rgee	(7)
	Precipit ation	C	Accumulated precipitation (mm/h)	4 Hours	1,000	JAXA GSMaP	Rgee	(8)
Anthropoge nic	Human density	C	Estimated human density (inh / sq-km)	Static	30	Facebook	website	(9)
	Agricult ure	B	Presence of agriculture	Static	30	Globeland 30 / Cropland	website	(4) & (10)
	Roads	B	Presence of roads	Static	vectorize d	Open Street Map	website	(11)
Light Intensity	Night	B	Binary variable (0 = day, 1 = night)	4 Hours	-	-	moonl it	(12)
	Moon illumina tion	C	Estimated moonlight illumination (lux)	4 Hours	-	-	moonl it	(12)

Note: The covariates in gray were combined into single proxies of water, human influence, and light intensity, respectively. Layers were either of the continuous type (C) or binary (B). Variables in *italics* were later disregarded due to convergence issues. Download methods in *typewriter* font refer to R packages. References: (1) DiMiceli et al. (2015) (2) Didan (2015) (3) Yamazaki et al. (2019) (4) Chen et al. (2015) (5) C. Schaaf (2015) (6) ESA (2023) (7) Muñoz-Sabater et al. (2021) (8) Kubota et al. (2020) (9) Facebook (2019) (10) Xiong et al. (2017) (11) OpenStreetMap Contributors (2024) (12) Śmielak (2023).Further details on covariate preparation and aggregation are provided in Appendix A3 and by Hofmann et al. (2021).

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260 STEP-SELECTION MODELS

261 We modeled dispersers' movement kernels and habitat selection using integrated step-selection
262 functions (iSSFs; Fortin et al. 2005; Avgar et al. 2016), following the procedures described by Fieberg
263 et al. (2021) and Muff et al. (2020). For this, we identified bursts of subsequent GPS locations where
264 the duration between two locations did not exceed 4 hours (± 15 minutes) or 8 hours (± 30 minutes,
265 for data recorded between 09:00 and 17:00). Within each burst, we converted locations into steps,
266 where a step represented the straight line segment between two consecutive locations (Turchin, 1998).
267 For each step, we computed the associated step length (sl, in meters) and relative turning angle (ta, in
268 radians). During this pre-processing, four coalitions dropped due to temporally irregular data, leaving
269 a total of 26 dispersing coalitions (12 female coalitions, 14 male coalitions) for further analyses. The
270 final dataset comprised 5,310 steps (204 ± 225 per coalition), which we further categorized into steps
271 collected during the wet season (15 October to 15 April) and steps collected during the dry season (15
272 April to 15 October), yielding 3,124 steps (59%) during the dry season and 2,186 steps (41%) during the
273 wet season. Because previous research revealed no differences in dispersal behaviour between males
274 and females, we pooled data across sexes for all analyses (Woodroffe et al., 2020).

275 We paired each *observed* step with a set of 100 *random* steps, generated by sampling turning angles
276 from a uniform distribution $U(-\pi, +\pi)$ and step lengths from a gamma distribution fitted to observed
277 steps (scale $\theta = 6341.96$ and shape $k = 0.39$). Together, an observed step and its associated 100 random
278 steps formed a stratum that received a unique identifier. Along each step, we extracted covariate
279 values from the underlying spatial layers and determined light conditions (Table 1). We opted for
280 covariate extraction *along* steps rather than *at their endpoints*, as we believed that environmental
281 conditions along steps were relevant in determining wild dog movement during dispersal (sensu
282 Hofmann et al., 2023). For continuous covariates, we computed average values along each step, for
283 categorical covariates the percentage cover of each category along the step. To model a decreasing

284 marginal impact of “distance-to-feature” variables, we included their square-root as predictors in the
 285 final models. To facilitate model convergence, we normalized extracted values to a range between zero
 286 and one.

287 To estimate movement and habitat parameters of interest, we applied the maximum likelihood
 288 procedure proposed by Muff, Signer, and Fieberg (2020), using the glmmTMB package (Brooks et al.,
 289 2017). To examine whether the effect of accounting for seasonality was modulated by the model
 290 structure, we employed two models informed by previous studies on wild dog dispersal (Hofmann et
 291 al., 2021, 2023). In the *simplistic model*, we exclusively included covariates relating to spatial features,
 292 alongside the movement descriptors (sl, log(sl), cos(ta)) that are mandatory to fit an iSSF (Avgar et al.,
 293 2016; Fieberg et al., 2021). This formula is akin to the models that are used with resistance-based
 294 approaches, which, however, preclude a mechanistic understanding of movement. In the *realistic*
 295 *model*, by contrast, we considered additional interactions between step descriptors, landscape
 296 characteristics, climate descriptors and light intensity, to account for the fact that wild dog movement
 297 during dispersal may be affected by such interactions too. In both models, we also included a stratum-
 298 specific intercept with a large, fixed variance (10^6 ; Muff, Signer, and Fieberg 2020) and used dispersing
 299 coalition ID to model random slopes for all main effects. To ensure comparability across the six
 300 configurations of seasonal dynamism (Figure 1), we applied the same model structures but varied the
 301 data used for model fitting (Figure 1a). In configuration SSS, we used static environmental data and
 302 pooled movement data across seasons. In configuration SMS, we used static environmental data, but
 303 we split the movement data by season. In configurations DSS and DSD, we used dynamic covariates
 304 but pooled movement data. In configurations DMS and DMD, we used dynamic covariates and split
 305 movement data by season. For each configuration, we fit both a simplistic and realistic model. To
 306 quantify how many random steps were necessary before model estimates stabilized (sensu Fieberg et
 307 al., 2021), we fitted each model with 5, 10, 25, 50, 75, and 100 random steps.

308 **VALIDATION**

309 We compared the predictive efficacy of each of the six configurations presented in Figure 1b using k-
310 fold cross-validation for case-control studies (Fortin et al., 2009). For this, we split movement data into
311 training and validation sets using an 80:20 ratio and fit the iSSF models described above. We then used
312 the obtained β -estimates to predict the probability of each random and observed step in the validation
313 set for being chosen. Depending on the configuration, we predicted step-probabilities using static or
314 dynamic covariates. Within each stratum, we assigned ranks 1-101 to each step based on predicted
315 probabilities and recorded the number of times the observed step was assigned each rank. Finally, we
316 computed Spearman's rank correlation between ranks and associated frequencies $r_{s,realized}$. The
317 better a model's predictive ability, the more negative $r_{s,realized}$ should be (i.e., the less often the
318 observed step should be assigned a low rank). For reference, we also computed Spearman's rank
319 correlation for randomized preferences ($r_{s,random}$), which we achieved by removing the observed step
320 from each stratum and identifying the rank of a randomly chosen step. We replicated this validation
321 procedure 100 times and employed ANOVA to compare the predictive efficacy of different
322 configurations.

323 **SIMULATIONS**

324 To assess differences in predicted connectivity upon increasing seasonal dynamism and model realism,
325 we ran dispersal simulations under the two most distinct configurations, i.e. the fully static (SSS, Figure
326 1) and fully dynamic (DMD, Figure 1) configurations, once using the simplistic model, once using the
327 realistic model. We simulated individuals starting from three distinct regions known to host viable wild
328 dog populations (Figure 2). The definition of these areas was somewhat arbitrary, albeit we deliberately
329 selected areas in the west and east of the Okavango Delta to examine potential influences of flooding
330 on connectivity (sensu Hofmann et al., 2024). We also selected a more independent location (Nxai Pan
331 National Park; Figure 2) that remained unaffected by the Okavango Delta's flood extent. To initiate

332 simulated dispersal trajectories, we randomly placed 1,000 starting points within each of these source
333 area, with starting dates that were equally distributed across the year. To simulate dispersal and obtain
334 connectivity maps under both configurations, we applied the simulation algorithm for iSSFs described
335 by Signer, Fieberg, and Avgar (2017) and employed by Hofmann et al. (2023). We applied the simulation
336 algorithm as follows. Originating from the starting point, we generated a set of 25 random steps by
337 sampling step lengths from the fitted gamma distribution and turning angles from a uniform
338 distribution. Along each random step, we extracted spatial covariates, computed relevant movement
339 metrics (sl, log(sl), cos(ta)), and assigned light conditions. We then employed the fitted iSSF model to
340 predict the probability of each step for being chosen, thus generating a “redistribution kernel” around
341 the current location of the simulated individual (Signer et al., 2017, 2024). Based on this kernel, we
342 probabilistically sampled one of the steps and updated the simulated individual’s position and time.
343 We repeated the procedure until a total of 2,000 steps were realized (~ 400 dispersal days).

344 Depending on the configuration, we employed different spatial covariates and preferences during the
345 simulation. For the SSS configuration, we omitted any seasonality and predicted connectivity using the
346 static set of covariates and fixed preferences (i.e., β -estimates). For the DMD configuration, on the
347 other hand, we updated covariates and preferences dynamically as the simulation progressed. That is,
348 after every simulated step, we generated a new covariate stack comprising spatial layers that best
349 represented environmental conditions at that particular point in time (Figure 3). For the purpose of
350 this comparison, we employed the set of covariates representing an average year. Similarly, we
351 predicted the redistribution kernel using β -estimates from the appropriate season. Finally, following
352 Hofmann et al. (2023) we obtained heatmaps under each configuration and computed difference maps
353 to highlight differences in resulting connectivity patterns.

354 RESULTS

355 Model performance was generally robust across all considered configurations. Nonetheless, due to
356 convergence issues, we removed the covariates NDVI, precipitation, and distance to pans from all
357 analyses. While this improved overall model convergence, some minor convergence issues remained,
358 as our validation procedure failed in 59 (i.e., 3%) out of 1,800 validation attempts. Since convergence
359 issues occurred randomly across different configurations, we deemed their exclusion of no concern.

360 STEP-SELECTION MODELS

361 Patterns of habitat selection and movement behavior were qualitatively similar, irrespective of
362 whether models were fit using static or dynamic covariates (Figure 4, Tables S3-S6) and irrespective of
363 the number of random steps considered (Figure S12). The most notable quantitative difference when
364 using static vs. dynamic covariates was that models fitted using dynamic covariates resulted in
365 narrower confidence intervals (Figure 4). Furthermore, effect sizes (i.e., β -estimates) of models fit using
366 static covariates were more pronounced than effect sizes from models that were fit using dynamic
367 covariates (Figure 4). Differences were most marked for $\beta_{DistanceToWater}$, which was estimated ≈ -0.8
368 using static covariates, but ≈ -0.3 when using dynamic covariates. Differences in β -estimates between
369 seasons were moderate and most pronounced when models were fit using static covariates. For
370 instance, when models were fit using static covariates, β_{Water} was ≈ -0.9 during the dry season but ≈ 0.2
371 during the wet season. This indicates avoidance of water during the dry season, yet selection for water
372 during the wet season. When dynamic covariates were used, by contrast, β_{Water} was ≈ -0.6 and ≈ -0.3 ,
373 respectively, indicating avoidance of water, irrespective of the season. Regardless of the type of
374 covariates used, $\beta_{sl:Dark}$ was markedly lower during the wet season (≈ -0.9) than during dry season
375 (≈ -0.1). This implies that dispersers generally avoided long steps during dark nights, with this tendency
376 being more pronounced in the wet season. Plots that aid with the interpretation of the models are
377 provided in Figure S13.

378 **VALIDATION**

379 Spearman's rank correlation coefficients (r) obtained from the validation procedure revealed that
380 predictions from the realistic model performed markedly better than those from the simplistic model
381 ($\bar{r}_{simplistic} = -0.5$, $\bar{r}_{realistic} = -0.9$, Figure 5). Irrespective of the employed model, Spearman's rank
382 correlation differed significantly depending on the amount of dynamism considered (ANOVA_{simplistic}:
383 $F(5, 593) = 26.45$, $p < 0.001$, ANOVA_{realistic}: $F(5, 594) = 7.14$, $p < 0.001$), albeit with moderate effect sizes.
384 Using the simplistic model, moving from a fully static (SSS) to a fully dynamic (DMD) configuration
385 decreased Spearman's rank correlation (i.e., increased the predictive performance) by 0.15 from -0.41
386 to -0.56. With the realistic model, conversely, moving from a fully static (SSS) to a fully dynamic (DMD)
387 configuration entailed a minor increase in r (i.e., a decrease in the predictive performance) by 0.01
388 from -0.89 to -0.88. Our hypothesis that accounting for seasonal dynamism results in better predictive
389 performance only held when using the simplistic model (Figure 5a), but not when employing the
390 realistic model (Figure 5b).

391 **SIMULATIONS**

392 Simulations under the most static (SSS) and dynamic (DMD) configurations revealed different
393 connectivity patterns (Figure 6). In the static configuration, simulated trajectories were more
394 concentrated, suggesting frequent movement across a few key habitats. In the dynamic configuration,
395 by contrast, simulated trajectories were more homogeneously distributed, suggesting that movement
396 was less restricted. When comparing the heatmaps generated under the simplistic and realistic models,
397 differences in predicted connectivity are imperceptible (Figure 6).

398 **DISCUSSION**

399 We developed and tested a conceptual framework for incorporating seasonality into connectivity
400 analysis at three stages: (1) when extracting spatial covariates for fitting the selection model, (2) when

specifying the selection model, and (3) when predicting from the fitted model. Using movement data from dispersing African wild dogs, we compared the predictive performance of six resulting configurations that varied in seasonal dynamism and biological realism. While seasonality had limited influence on estimated patterns of habitat selection and movement behavior, and only marginally improved predictive performance overall, the realistic model alone achieved such high predictive efficacy that increasing seasonal dynamism only marginally improved predictions. Despite this, our simulations showed that seasonal dynamism can lead to contrasting connectivity patterns: Static configurations produced clumped connectivity around a few hot spots, whereas dynamic configurations yielded more homogeneously distributed connectivity across the landscape.

Our validation procedure revealed only moderate improvements in our ability to predict dispersal movements when increasing the degree of seasonal dynamism. Given the system's extreme seasonal variability, this was somewhat surprising. We suspect that the absence of a pronounced improvement can be traced back to multiple factors. Firstly, we focused our analysis on dispersing individuals, which cover large distances in search of potential mates and a suitable territory (Cozzi et al., 2020; McNutt, 1996). In our case, the average 4-hourly step length was 2.5 km, implying that dispersers cross and sample a large variety of habitats within short time. The spatial scale of the dispersal process may thus misalign with the spatial scale at which seasonality affects connectivity. A second explanation could emerge from the fact that dispersers prioritize finding unoccupied territories and areas with low competition, which are unlikely to represent optimal habitat conditions, over selecting specific habitat types (e.g., Creel, 2001; Creel & Creel, 1996). Dispersers' habitat selection may therefore be more strongly influenced by the spatial organization of territorial, resident conspecifics and only little affected by the landscape (Cozzi et al., 2018) and its seasonally changing characteristics. This would also explain the relatively weak selection or avoidance of environmental characteristics exhibited in Figure 4. Finally, the African wild dog is a generalist species that occupies a broad variety of habitats

(Woodroffe, 2011). A certain tolerance towards changing environmental conditions can therefore be expected and may explain why seasonal differences were faint. This holds particularly true for dispersers, which are usually more tolerant towards unfavorable habitat conditions as they spend little time within the same area (Elliot et al., 2014; O'Neill et al., 2020).

Even though incorporating seasonality provided little benefit in terms of predictive performance in our system, we stress that adopting our conceptual framework will facilitate greater clarity and rigor in connectivity studies overall. By explicitly defining where and how seasonality enters the modeling pipeline, our framework forces researchers to be more specific about the type of seasonal variation their study accounts for, reducing ambiguity in how seasonal effects are represented. Additionally, it prompts a more deliberate consideration of the pathways through which seasonality can influence connectivity, helping to identify whether its effects primarily stem from environmental variability, shifts in species behavior, or both. Our approach not only enhances transparency and comparability across studies but also fosters a more realistic and mechanistic understanding of movement, ultimately improving the ecological relevance of connectivity analyses.

Compared to the improvements achieved by increasing seasonal dynamism, moving from the simplistic to a realistic model more significantly benefited predictions. Through the realistic model, we could represent several important aspects of African wild dog behavior, such as reduced movement during dark nights (Cozzi, 2013; Cozzi et al., 2012) and during times of high ambient temperature (Rabaiotti & Woodroffe, 2019). We could also model that dispersers' movement behavior depended on habitat conditions, as evidenced by shorter steps in areas covered by water (Hofmann et al., 2023). The ability of encapsulating and replicating dispersal movements in such realistic manner is unique to iSSF-based simulations and cannot be achieved using resistance-based models. This is also a testimony to step-selection functions as a powerful means of conducting both inference and prediction of movement and connectivity (Forrest et al., 2024; Hofmann et al., 2023). While many of the additionally encapsulated

processes are not directly linked to landscape connectivity, incorporating them helps control for key behavioral and environmental influences that might otherwise obscure critical movement patterns. In our case, this versatility elevated the predictive power of our dispersal model to a level where the added benefit of considering seasonal dynamism was negligible.

Comparisons of connectivity under the most static (SSS) and most dynamic (DMD) configurations, revealed that connectivity was concentrated on a few major dispersal hotspots under the most static configuration, but uniformly distributed across the entire landscape under the most dynamic configuration. This corroborates previous research showing that static connectivity analyses tend to underestimate connectivity, as seasonal stepping stones or corridors are missed (Martensen et al., 2017). Seasonally available dispersal habitats periodically allow dispersers to access areas that would otherwise remain inaccessible. In our case, such stepping stones emerged in two ways: through changes in the landscape that reshaped the distribution of preferred habitats, and through seasonal shifts in the dispersers' habitat preferences. For instance, the pulse-like flood dynamics of the Okavango Delta seasonally reveal vital dispersal habitats through which individuals access otherwise inundated regions. Similarly, seasonal differences in the species willingness to move far distances at night modulate the likelihood of dispersal between adjacent protected areas. Jointly, seasonal landscapes and changing preferences resulted in a more balanced mosaic of connectivity. Note, however, that the fact that connectivity was more evenly distributed across the year does not exclude the possibility of certain areas experiencing lower connectivity seasonally. As already shown by Osipova et al. (2019), a static representation may result in both an over- and under-estimation of short-term connectivity, depending on the area and season. For conservation planning, this implies the need to protect landscapes more broadly, as areas that provide little connectivity during some season may be critical during others. A dynamic take at connectivity therefore improves our understanding of

ecological processes in dynamic landscapes and may help to identify otherwise overlooked dispersal hotspots (Zeller, Lewison, et al., 2020).

Our results showed only marginal improvements in predictive efficacy with increased seasonal dynamism; however, modest differences in step-level predictions translated into pronounced differences in emergent connectivity patterns. While this may seem contradictory, it likely arises from our validation approach, which was constrained to the step level due to a fundamental limitation in validating seasonal connectivity from an iSSF. In dynamic landscapes with shifting preferences, connectivity varies both spatially and temporally, requiring validation at each timestamp in the dataset. Since our landscape changed every four hours (i.e., with each step), a new connectivity map would need to be generated around each validation step, effectively reducing validation to the step level. Given this constraint, we deemed the application of k-fold cross-validation using Spearman's rank correlation at the step level appropriate. Note, however, that Spearman's rank correlation is influenced by the number of steps per stratum, making it unsuitable for comparing predictive efficacy across studies (Figure S14). For static connectivity analyses, alternative validation techniques exist, which typically compare connectivity at observed vs. random locations (Brennan et al., 2020; Finerty et al., 2023; McClure et al., 2016; Zeller et al., 2018). Developing similar validation approaches beyond the step level for dynamic connectivity analyses remains a challenge for future research.

Increasing seasonal dynamism to model dynamic connectivity comes at significant costs, both in terms of data requirements and computational challenges. To represent seasonal changes across the landscape, spatial layers need to be downloaded and processed at an appropriate seasonal interval. This is time-consuming and implies a substantial increase in the data-volume to be handled. Seasonal products are less commonly available, potentially necessitating custom remote-sensing analyses (e.g., Appendix A4). Moreover, seasonal layers are often plagued by noise and missing data, especially during seasons with extensive cloud cover. Extracting covariates from seasonal layers poses a further

challenge, as covariate data can no longer be extracted from single layers but need to be extracted from those layers that best represent environmental conditions at the time of the observed movement. This applies to the extraction of covariates for both model fitting (Figure 1, stage 1) and prediction (Figure 1, stage 3). Finally, splitting species data by season to model seasonal preferences reduces the amount of data remaining per season, and may lead to changes in parameter estimates, wider confidence intervals, inconsistent effect sizes and potential convergence issues. In our case, the moderate improvement in predictive ability did not justify these additional efforts and complications. The trade-offs between increased realism and practical applicability must therefore be carefully considered and aligned with the research goal, as greater complexity does not always lead to better insights and may even obscure key ecological patterns (Puy & Saltelli, 2022).

Whether incorporating seasonal dynamism improves connectivity predictions likely depends on several key factors. Species traits play a central role: habitat specialists may respond more strongly to seasonal environmental changes than generalists, whose broader tolerances buffer them against such variability. Similarly, animals with limited mobility may be more sensitive to fine-scale seasonal changes than long-distance dispersers, for whom connectivity patterns are shaped by broader-scale landscape structure. Landscape characteristics also matter: dynamic connectivity may be more relevant in heterogeneous, mosaic-like environments where seasonal shifts significantly alter movement corridors, compared to more homogeneous landscapes.

In conclusion, we provided a conceptual framework to integrate seasonality into connectivity analyses and investigated the role of seasonal dynamism in modeling dispersal and connectivity. While increasing the level of dynamism led to only moderate improvements in predictive performance, it had a substantial impact on inferred connectivity patterns, resulting in more evenly distributed connectivity when seasonality was considered. This suggests that seasonal fluctuations can shape connectivity in ways that may not be immediately evident from overall predictive accuracy alone. However, the added

520 effort of incorporating seasonal landscapes and seasonal species preferences must be carefully
521 weighed against its practical benefits. Beyond the effects of seasonal dynamism, our findings highlight
522 that model structure plays a more crucial role in shaping movement predictions. In our case, a
523 biologically realistic and mechanistic model, which explicitly accounted for species-specific movement
524 constraints, outperformed a simplistic model, akin of those typically used with resistance-based
525 approaches. In fact, a realistic model obliterated any improvement of accounting for seasonality by
526 elevating predictive efficacy to a very high level. While this result may be specific to our study system,
527 it underscores the broader importance of selecting an appropriate movement model when studying
528 connectivity and it attributes a large potential to iSSF-based simulations for studying connectivity.

529 **AUTHORS' CONTRIBUTIONS**

530 D.D.H. and G.C. conceived the study and designed methodology; D.D.H., G.C., D.M.B, and J.W.M.
531 collected the data; D.D.H. analysed the data; G.C. assisted with the modeling framework; A.O.
532 contributed to conceptual discussions and provided feedback during the development of the study;
533 D.D.H. and G.C. wrote the first draft of the manuscript and all authors contributed to the drafts at
534 several stages and gave final approval for publication.

535 **DATA AVAILABILITY**

536 Access to R-scripts to replicate our analyses will be provided through an online repository at the time
537 of publication.

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543 improved the manuscript.

544 **IMAGE CAPTIONS**

545 Figure 1: Graphical abstract illustrating how seasonal dynamism and model realism can enter
546 connectivity studies. (a) Different stages at which seasonality can be incorporated in studies of
547 dispersal and landscape connectivity. Stage 1: When fitting a model, the environment can be
548 represented by a static set of covariate layers, thus ignoring seasonality, or by a dynamic stacks of
549 covariate layers that capture how the landscape changes over time. Stage 2: Once covariates are
550 extracted, one also needs to decide whether to parameterize a single-season model, assuming fixed
551 species preferences across the year, or a multi-season model that accounts for seasonal differences in
552 the species habitat selection and movement behaviour. Stage 3: When predicting connectivity from a
553 fitted model, one can use a static set of environmental covariates or employ seasonally update
554 covariates. Via combination, this results in six different configurations that differ in terms of the degree
555 of seasonality they encapsulate (see colored codes). (b) We furthermore examined the importance of
556 moving from simplistic models (akin to those used in resistance-based connectivity models) to realistic
557 models (akin to those used with individual-based simulations), where simplistic models are limited to
558 capturing an animal's habitat selection, whereas realistic models also account for the animal's
559 movement kernel and potential interactions. (c) Our hypotheses were that increasing the degree of
560 seasonal dynamism would positively affect predictive efficacy, and that a realistic model would
561 outperform a simplistic one. Dots are showing hypothetical predictive efficacy values and associated
562 standard deviation for replicated simulations. This is a stylized figure and the hypothesized relationship
563 may differ for other study systems.

564 Figure 2: (a) Study area within which GPS data on dispersing African wild dogs were collected. Dispersal
565 trajectories are displayed in dark gray. The study area encompassed parts of the Okavango Delta in
566 northern Botswana, a highly dynamic, flood-pulse-driven ecosystem. The entire study area undergoes
567 substantial seasonal changes, as can be seen from two satellite images taken during peak dry season
568 (b1) and peak rainy season (b2). Notably, the flood of the Okavango Delta reaches its maximum extent
569 during peak dry season between August and September. Source areas from which dispersers were
570 simulated are highlighted in orange. Satellite images in panels b1 and b2 are from the MODIS Terra
571 sensor and were provided through NASA Worldview Snapshots. Sun and cloud icons are original
572 artwork created by the authors in Inkscape.

573 Figure 3: Schematic illustration of the dispersal simulation framework with dynamic covariates. As the
574 simulation progresses and the simulated individual progresses through seasons, the underlying spatial
575 covariates (represented by the stacked layers) are updated through time. Update frequency varies
576 among covariates, ranging from a few hours (e.g., temperature) to multiple months (e.g., shrub cover).
577 Each gray block represents a single covariate layer and the time period for which it is active. At each
578 step, a redistribution kernel is constructed from the animal's current location, representing a set of
579 potential movement steps and their predicted selection probabilities. Redistribution kernels are
580 generated by proposing 25 random steps and applying the parameterized integrated step-selection
581 (iSSF) model to estimate the relative probability of each step being chosen. One location is then
582 probabilistically sampled from this kernel, and the animal's position and time are updated accordingly.
583 This procedure is repeated until 2,000 steps (approximately 400 dispersal days) are simulated. The
584 figure illustrates how covariates with different temporal resolutions are dynamically assembled to
585 represent environmental conditions at each simulated point in time. All icons and silhouettes are
586 original artwork created by the authors using Inkscape and Microsoft PowerPoint.

Figure 4: β -estimates from the integrated step-selection models, grouped by movement kernel, habitat-selection function, and their interaction. We either fit models using a simplistic model (without interactions) or a realistic model that comprised several interaction terms that rendered how movement behavior depended on environmental conditions. Furthermore, we distinguished between models fit using static and dynamic covariates. However, only the underlined covariates differed between the static and dynamic configurations, as covariates were either represented by a single layer (static), or a stack of layers (dynamic). Furthermore, data were either pooled across seasons (yellow bars) or split into dry (green bars) and wet (blue bars) season. Results are shown for replicates that used 25 random steps, as all associated models converged. The number of random steps did not appear to influence model estimates (Figure S12). The horizontal ticks indicate the point estimates, and the vertical segments represent the 90%, 95%, and 99% confidence-intervals of the respective β -coefficients.

Figure 5: (a) Boxplots of Spearman's rank correlation across different configurations of dynamism that range from entirely static (SSS) to fully dynamic (DMD). The more negative Spearman's rank correlation, the better is the predictive performance under the respective configuration. Correlations were computed for 100 replicates. Note that the y-axis is inverted to match our expectation of increasing performance as dynamism increases. The boxplots in light gray represent Spearman's rank correlation coefficients under a null model and serve for comparison. (b) Difference in Spearman's rank correlation when moving from one configuration to another. Values < 0 (blue) indicate an increase in predictive performance, whereas values > 0 (orange) indicate a decrease in predictive performance.

Figure 6: (a-d) Heatmaps of the dispersal simulations generated under the most static (SSS) and most dynamic (DMD) configurations, using both a simplistic and a realistic model, as well as pixel-wise difference maps highlighting discrepancies in the number of simulated dispersers traversing a certain region between these modeling approaches. Originating from each of the three areas highlighted in

611 the maps, 1,000 dispersers were simulated, moving for a total duration of 2,000 steps (~400 dispersal
612 days). In the most static configuration (SSS), seasonality was ignored across all stages of the
613 connectivity modeling pipeline, from extracting covariates for model fitting (stage 1), in the model itself
614 (stage 2), and when predicting from the model (stage 3). In the most dynamic configuration (DMD), on
615 the other hand, seasonality was incorporated across all those three stages. Furthermore, in the
616 simplistic model, only the species' habitat selection was considered, whereas in the realistic model also
617 the species' movement kernel, and potential interactions between habitat selection and movement
618 behaviour were included.

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